

Charles Dsouza

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PROFESSIONAL SUMMARY

Aspiring Software Engineer with full-stack development experience and a strong foundation in data structures, algorithms, and software engineering practices. Skilled in Python, JavaScript, C++, and Java, with experience building and deploying end-to-end applications using React.js, Node.js, and REST APIs. Seeking a Software Engineer role to design and build scalable, maintainable software.

SKILLS

Frontend: HTML5, CSS3, JavaScript (ES6+), React.js, Next.js (App Router), Tailwind CSS, TypeScript

Backend: Node.js, Express.js, FastAPI, REST API design and implementation, SQLAlchemy

Databases: MongoDB, MySQL, PostgreSQL, ChromaDB (Vector DB)

Languages: Python, C++, Java

Tools & Platforms: Git, GitHub, Vercel, Render, VS Code, npm, Firebase, pytest, LangSmith

AI / ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, Pandas, NumPy, LangChain, LangGraph, RAG, RAGAS

EDUCATION

NMIMS, Navi Mumbai — B.Tech, Computer Engineering

2022 – 2026

CGPA: 3.13 / 4.0 ($\approx 7.83 / 10$)

EXPERIENCE

Nair Super Speciality Eye Hospital

January–June 2026

Data Analyst and Web Development Intern

- Developed full-stack healthcare web applications and interactive clinical dashboards (IPD, reception, and feedback analysis) using Node.js, Express, and PostgreSQL to streamline hospital data tracking.
- Built an AI-powered medical chatbot leveraging a Retrieval-Augmented Generation (RAG) pipeline and Transformer models, and designed custom digital workflows to improve patient communication and health awareness.

CodSoft

June 2024(1 month)

Full Stack Web Development Intern

- Built responsive, production-ready web pages using HTML5, CSS3, JavaScript, and React.js within an Agile team environment.
- Managed codebase via Git/GitHub, taking end-to-end ownership of UI components from design through deployment.

PROJECTS

EmpathixAI — A Full-Stack Agentic RAG Medical Assistant [\[Live Demo\]](#)

- Built EmpathixAI (React, FastAPI, LangChain, ChromaDB):** Developed a full-stack RAG medical chatbot with an agentic triage workflow to safely route emergency queries and minimize hallucination risks.
- Engineered Multimodal RAG & Testing:** Implemented document/image analysis via a vision LLM, quantified system performance using a RAGAS evaluation harness, and built a robust pytest suite for CI testing.
- Delivered Production-Ready App:** Deployed a fully localized (English-Hindi) application on Vercel and Render, featuring Firebase authentication and LangSmith request tracing.

AI Powered Exercise Form Classification and Correction [\[Github\]](#)

- Built an AI-powered exercise form classifier in Python (TensorFlow, MediaPipe, PySide6) to analyze bicep curl and shoulder press posture in real time.
- Implemented an Attention-Augmented BiLSTM network and OpenCV pose estimation to evaluate 30-frame temporal sequences using joint-angle biomechanical validation.
- Delivered live 30 FPS feedback, automated rep-counting, and correction alerts via an interactive desktop GUI with performance analytics.

EXTRACURRICULAR ACTIVITIES

Robin Hood Army — Volunteer

42 Days

- Contributed to hunger-relief drives and mentored underprivileged children, building leadership, communication, and teamwork skills.